

Alison Seongjee Park

Computational oncology | Multiple myeloma | Multi-omics | Single-cell analysis

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Bioinformatician in computational oncology focused on multiple myeloma, multi-omics integration, and single-cell analysis. Recent work spans a 389-patient subtype study, 185-sample immune-profiling analyses, and SETD8 research that culminated in a Journal of Medicinal Chemistry paper.

SELECTED EXPERIENCE

Bioinformatician

December 2023 to Present

Icahn School of Medicine at Mount Sinai

Multiple Myeloma Computational Research

- Integrated bulk RNA-seq, single-cell RNA-seq, and clinical data using SNF, cNMF, MOVICS, MOFA, and integrAO to stratify newly diagnosed multiple-myeloma patients.
- Built patient-similarity networks that refined disease classification beyond traditional staging systems and surfaced immune-based subtypes with distinct outcomes.
- Performed single-cell analysis in Scanpy and pySCENIC to identify depleted antigen-presenting-cell and IFN-stimulated CD4 T-cell populations in high-risk disease.

Research Assistant

August 2018 to November 2021

Icahn School of Medicine at Mount Sinai

Cancer Epigenetics Laboratory

- Advanced from volunteer trainee into an owned SETD8 inhibitor project while continuing SOX11 and E2F1/selinexor programs in multiple myeloma and mantle cell lymphoma.
- Screened collaborator-derived compounds in cell lines, measured efficacy and toxicity with flow cytometry and cell-culture assays, and validated response with qPCR, Western blotting, cell-separation assays, and CRISPR knockout studies.
- Ran xenograft studies in immunocompromised mice against saline placebo and partnered with pathologists on IHC/imaging review of harvested tumors; related E2F1/selinexor work produced an ASH abstract and SETD8 work matured into a 2026 JMC publication.

Research Trainee and Early Internships Mount Sinai and Northwell Health

2015 to 2018

Cancer biology foundations

- Built early cancer-biology foundations through internships in chronic lymphocytic leukemia, chemo-resistant bladder cancer, and SOX11-focused mantle-cell lymphoma.
- Learned project-building from senior scientists and a postdoctoral mentor through compound screening, cell culture, flow cytometry, PCR/qPCR, Western blotting, lentiviral transduction, mouse-model workflows, ChIP-seq exposure, and scientific writing.
- Shadowed clinical decision-making in hematologic malignancy to connect bench assays with how clinicians diagnose, monitor, and interpret disease.

EDUCATION

CUNY School of Medicine

MD Candidate | August 2021 to January 2023

Sophie Davis / The City College of New York

BS Biomedical Sciences, Cum Laude | August 2017 to May 2021

Flatiron School

Data Science Bootcamp | February 2023 to June 2023

TECHNICAL SKILLS

Programming and analysis: Python, R, SQL, Scanpy, Seurat, pySCENIC, survival analysis

Genomics and single-cell work: bulk RNA-seq, single-cell RNA-seq, DESeq2, GSEA, InferCNV, CellphoneDB, CyTOF integration

Modeling methods: SNF, cNMF, MOVICS, MOFA, integrAO, clustering, feature engineering

Workflow and infrastructure: Git/GitHub, shell scripting, HPC environments, reproducible workflows, notebook-based ML analysis